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Review

Advance care planning among stroke patients: A scoping review

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ABSTRACT

Objective: This study aimed to describe the implementation of advance care planning (ACP) in patients with stroke.
Methods: A scoping review was conducted according to the Arksey and O'Malley framework. Accordingly, we searched for articles published in ScienceDirect, PubMed, Scopus, ProQuest, and Medline. The inclusion criteria were original research on ACP among adult or elderly stroke patients, hospital- or community-based studies, and those published in English or Indonesian. Qualitative analysis was then used to identify keywords, categories, and themes.
Results: Among the 1,050 articles identified, only 8 satisfied the inclusion criteria, indicating that limited studies were available for ACP interventions among stroke patients. Four themes were ultimately identified: the strategy implementation of ACP, the challenges of ACP implementation, the benefits of ACP, and the factors influencing ACP. Strategy ACP implementation included the ACP method, the communication style, participation role, media used, and ACP documentation. Lack of information about ACP, facilities, incapacitated patients, and reluctance to implement have become challenges to implementation in stroke. The benefits of ACP, particularly the outcomes of the ACP program, included awareness of ACP engagement, patients can express their wishes, and increased advance directive (AD) documentation. The factors influencing ACP included patient condition, knowledge, and appropriate ACP methods.
Conclusion: ACP has not been widely used when providing interventions for stroke patients. The implementation of ACP in stroke patients is full of challenges due to various patient conditions. Effective strategies are needed to improve ACP in stroke patients. It is necessary to develop an ACP for stroke patients' method that involves interprofessional collaboration, and studies on ACP in stroke patients need to be further conducted through interprofessional collaboration.
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What is known?

- Advance care planning (ACP) is a process that involves discussing personal values, life goals, and preferences related to their future health to support adults regardless of health condition.
- Some studies on ACP among stroke patients have found that it improved their physical and psychosocial health.

What is new?

- ACP faced several challenges in stroke patients because of their varying conditions, and there was a need for effective strategies to enhance ACP among stroke patients.
- Despite its limited implementation in stroke care, the use of ACP proved effective in enhancing documentation and promoting patient engagement in ACP discussions.

1. Introduction

Stroke remains the second leading cause of death and the third leading cause of disabilities worldwide [1]. The prevalence of stroke

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in 2019 around the globe has reached 101 million, among which 12.2 million were new cases [2]. Stroke occurs when blood flow to the brain is interrupted due to a blood clot or rupture of a blood vessel in the brain, causing death or severe disabilities [3]. The number of patients who have died due to stroke has reached 6.5 million, with over 143 million stroke patients losing their health and becoming disabled [2]. Among the Southeast Asian countries, in 2015, Indonesia had the highest death rate due to stroke, with 193 deaths having occurred per 100,000 people per year [4]. According to the Indonesian Basic Health Research, the prevalence of stroke within the country had reached 10.9 % as of 2018 [5].

Stroke causes various complications and increases physical and psychological problems [6]. Patients who suffer a stroke often present with cognitive disorders, paralysis, unstable emotions, anxiety, depression, disability, socioeconomic problems, family caregiver burden, and spiritual distress [3,7–9]. The disabilities that happen due to weakness and paralysis deteriorate the patients' quality of life [7], thereby causing anxiety and low self-esteem [10,11]. Stroke patients may experience changes in their condition such that their health or consciousness deteriorates, preventing them from making decisions regarding their future health care [12,13]. In such conditions, the family will be responsible for making decisions [14]. This often becomes a burden for the family, which complicates the decision-making process [14] and frequently creates conflict among them [15], which complicates the decision-making process for stroke patients [16].

Stroke patients often lack information about stroke and its treatment and are unable to communicate feelings, emotions, and their preferences for health care, which can be problematic for them [1]. Hence, appropriate interventions are necessary to address these problems. Advance care planning (ACP), a process that helps adults of any health level understand and share their values, life goals, and preferences related to their future health care [17] may help patients communicate what treatments they need. One study found that ACP helped patients prepare for future decisions when they became ill and could no longer communicate their wishes [18]. ACP has been implemented in several Asian countries and worldwide using various models. In Asia, ACP is defined as a process through which the objective of end-of-life (EoL) care among health professionals is discussed and communicated with their patients and families [19].

Systematic or scoping reviews of ACP interventions have found that they can reduce stress among caregivers, increase the satisfaction of patients and their families, facilitate the documentation of the advance directives (ADs), improve patients' knowledge, and promote self-efficacy [20–22]. Moreover, evidence shows that ACP helps stroke patients receive appropriate treatment according to their needs due to their disabilities and risk of death [23], improves their physical and psychosocial health [24], reduces confusion, anxiety, and family conflict during decision-making [16,19], and decreases family caregiver burden [25].

A systematic review on ACP in patients with neurodegenerative disorders showed that 66 %, 13 %, 9 %, 4 %, and 2 % of ACP had been conducted in patients with dementia, motor neuron disease (amyotrophic lateral sclerosis), brain tumors, Parkinson's disease, multiple sclerosis, and Duchenne muscular dystrophy [20]. The authors of the mentioned review had searched several databases and found that studies related to ACP interventions on stroke patients were quite limited, even in European, American, Asian countries, and Australia. The novelty of this scoping review lies in its focus on illustrating the implementation of ACP among stroke patients, to support health professionals in delivering appropriate ACP interventions for this specific population. Thus, this scoping review aimed to describe the implementation of ACP in stroke patients through a scoping review of available literature.

2. Methods

2.1. Review design

A scoping review was performed using the framework recommended by Arksey and O'Malley [26]. This scoping review was conducted using the PRISMA-ScR [27]. The protocol for this scoping review has been registered with the Open Science Framework (OSF), accessed through https://doi.org/10.17605/OSF.IO/QW87J.

2.2. Identifying the research problem

This study addresses two main questions: 1) how ACP is implemented among stroke patients? and 2) what the outcomes of ACP among stroke patients are?

2.3. Literature search

A literature search was conducted based on the participants, concept, and context (PCC) framework recommended by the Joanna Briggs Institute [28]. Participants were adults or elderly patients with stroke; the concept was ACP; and the context was hospital or community.

The inclusion criteria in this research were as follows: 1) original research articles about ACP on stroke patients; 2) adults and elderly patients diagnosed with stroke; 3) hospital or community-based research; and 4) articles written in English or the Indonesian language. The exclusion criteria were as follows: 1) young and teenage participants; 2) studies on ACP among patients with neurological disorders other than stroke; and 3) articles containing only the abstract instead of full text, opinion-based articles, protocol studies, or review articles.

A scoping search was performed on five databases, ScienceDirect, PubMed, Scopus, ProQuest, and Medline, to identify studies published until March 2024. The search on PubMed was performed using the PCC framework and Medical Subject Headings. The literature search also employed other references, such as Google Scholar. Articles were identified using the Boolean operators "AND" and "OR" to combine the keywords. The keywords were "stroke/cerebrovascular accident/brain vascular accident," and "advance care planning/advance health care planning/advance medical planning" (Appendix A).

2.4. Selection of sources

The first screening was executed using the Rayyan software to eliminate data duplication. Two independent reviewers conducted further screening to select articles based on titles and abstracts, followed by a full-text review based on the inclusion criterion, performed independently. Any incongruities between the reviewers were resolved through discussions.

Based on the literature search conducted, a total of 1,136 articles were identified. Eighty-seven duplicate articles were removed using Rayyan software, leaving 1,050 articles. The remaining articles were screened based on the inclusion and exclusion criteria covering the titles, abstracts, and full texts. Screening of the titles revealed that 1,029 articles had inappropriate titles, leaving 21 articles. Screening of the abstracts revealed that six articles did not match the publication type, topic, and population, leaving 14 articles. The last screening was conducted on the full texts of the articles, among which eight were appropriate for analysis. The literature screening process is shown in Fig. 1.

2.5. Data extraction and data analysis

One author performed the data extraction, which another author verified. Data were entered into Microsoft Excel and presented in the form of tables. The tables show some information, which includes 1) author(s), year of publication, and country; 2) study design, sample, and setting; 3) intervener; 4) ACP procedure; 5) instrument; 6) outcome; 7) result; 8) strength/limitation. Data were analyzed qualitatively through keywords, categories, and themes using Braun and Clarke's thematic analysis approach [29] to identify patterns and themes emerging from the extracted data to formulate the result.

3. Results

There were eight articles from this study, and the characteristics are described in Appendix B. Stroke patients had a variety of clinical characteristics, both in terms of the type of stroke and the severity of the stroke they experienced. This study involved patients with various types of stroke, including transient ischemic attack (TIA), acute ischemic stroke, and intracerebral hemorrhage, as well as varying degrees of stroke severity: mild, moderate, and severe [30–33]. The ACP intervention providers in this study were physicians [30–33], medical providers [34,35], and clinical staff [36]. The majority of the research was conducted in the USA [30,31,34,35,37], while the rest was conducted in Germany [32], the Netherlands [33], and Australia [36].

Based on the thematic analysis of eight articles, four themes were identified, namely 1) implementation strategy of ACP in stroke; 2) ACP implementation challenges in stroke; 3) benefits of ACP; and 4) factors influencing ACP.

3.1. Theme 1: implementation strategy of ACP in stroke

The ACP approach was identified as one of the implementation strategies in six of the reviewed articles. One article explicitly reported using the PREPARE program to facilitate ACP [35], while the others described ACP implementation through structured discussions or the completion of ACP documentation [31–34,37]. These discussions were conducted formally [34,36] or informally [30]. Six studies explored discussion topics relevant to ACP among stroke patients, including personal values, patient wishes, treatment preferences, decisions regarding treatment limitations, patient concerns, and preferred place of death [30,32,34,36,37]. Notably, one study addressed the risk of stroke recurrence as a discussion topic [31], which was not reported in the other studies.

Eight articles described the individuals involved in ACP discussions, each fulfilling specific roles. Physicians [30–33], healthcare providers [34,35], notaries [32], family members [30,32], proxies or surrogates [30,33,34], and patients themselves [30–37] were identified as key participants in the ACP process. Physicians and healthcare providers played key roles in providing information about the patient's condition, facilitating discussions regarding the patient's preferences with the patient and their family [33], and participating in decision-making processes [34]. Patients contributed by expressing their preferences [36], communicating their wishes to family members and healthcare providers [31,34], and making decisions regarding their care [33,36]. Surrogates and family members were involved in expressing the patient's wishes when the patient was no longer able to do so [33], making healthcare decisions on the patient's behalf [31], participating in discussions [37], and serving as witnesses in the completion of AD documents. Media also served as an important tool in supporting ACP implementation strategies. Some studies explicitly mentioned using media such as videos, pamphlets, AD documents [35], and forms [34,36] to facilitate ACP discussions. The final strategy, discussed in three articles, was the availability of ACP documentation, including medical records [36] and AD documents [31], as well as the presence of witnesses—such as a notary, physician, or family member—to reinforce the legal and ethical validity of the documents [32].

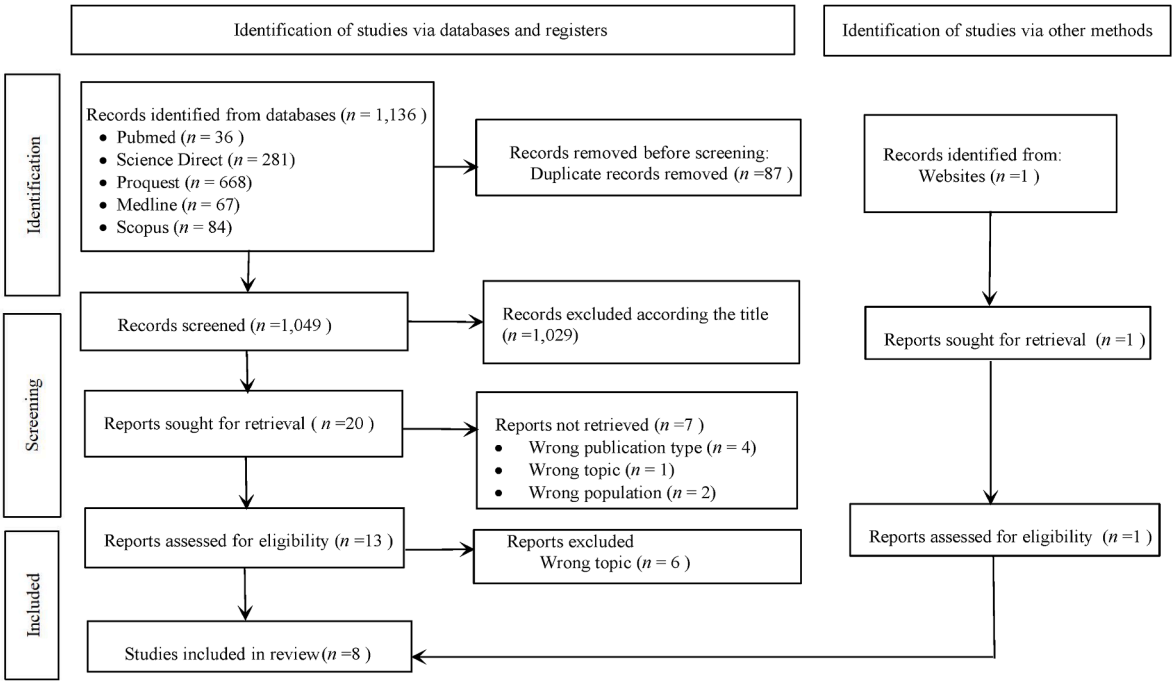


Fig. 1. Flow diagram of literature screening.

3.2. Theme 2: ACP implementation challenges in stroke

Four of the reviewed articles identified challenges related to the implementation of ACP. Skolarus et al. reported a low prevalence of ADs among stroke patients [34]. De Kort et al. found that patients and their families were generally unfamiliar with ACP, and there was a low level of agreement between patients' preferences and those of their surrogates [33]. Two articles highlighted facility-related barriers, including the limited availability of surrogates [33] and the use of certain technologies—such as telehealth—which posed additional challenges in facilitating ACP discussions [36]. Another significant challenge involved managing patients with compromised cognitive or communicative abilities. Stroke patients often experience varying degrees of impairment, including unconsciousness, aphasia, and cognitive deficits, which can hinder meaningful participation in ACP discussions [33]. Furthermore, patient reluctance to engage in ACP conversations and the failure to complete advance directive documents were persistent barriers to effective ACP implementation [32,33].

3.3. Themes 3: benefits of ACP

Two articles discussed the benefits of ACP from the perspective of patient communication. Patients who received ACP interventions could express their preferences regarding life-sustaining treatments and communicate their values concerning future healthcare decisions [36]. Additionally, these patients demonstrated an increased interest in participating in further ACP conversations [31]. Three studies highlighted the benefits of ACP in aligning care with patient preferences. These included patients' desires to discontinue treatment in cases of severe pain or disability, or to refuse specific treatments altogether [32], earlier transitions to comfort measures only (CMO) [30], and the application of treatment restrictions [33]. Other benefits of ACP were related to improvements in AD documentation. Two studies reported that ACP interventions led to higher rates of documented ADs. Malpass noted increased documentation in patients who participated in face-to-face ACP sessions [36]. At the same time, Skolarus et al. found greater AD documentation and a higher completion rate of ACP elements in the group that used the PREPARE program [35].

3.4. Theme 4: factors influencing ACP

Several factors influencing the implementation of ACP were identified in four articles. The first factor was patient condition, which included variables such as age, physical disabilities, activity limitations, comorbidities, and other health-related issues [31,37]. Patient knowledge regarding ACP also played an important role. This included prior discussions about ACP [31,37], conversations addressing the risk of stroke recurrence [31], and general unfamiliarity with ACP [31,37], all of which contributed to varying levels of patient understanding and engagement. The use of appropriate intervention strategies was another influencing factor. For instance, the application of the PREPARE program [35] and conducting face-to-face ACP sessions [36] were both associated with improved engagement and outcomes in ACP implementation.

4. Discussion

Only eight articles were found based on the study conducted, indicating that research on ACP in stroke patients is still minimal. The current scoping review aimed to investigate the implementation of ACP among stroke patients. Four out of the eight articles involved patients with all types of strokes—transient ischemic attacks, ischemic stroke, and intracerebral hemorrhage (ICH)—with varying degrees of severity, ranging from mild to severe (median NIHSS score, range: 4–19).

These studies found that patients with limitations in activities of daily living (ADL), those with severe stroke, or those lacking decision-making capacity tend to involve family members/surrogates in the decision-making process [31,33,36]. The type and severity of the stroke, as well as the affected areas of the brain, significantly determines the clinical condition of the patient. Patients with mild to moderate stroke generally retain sufficient consciousness and communication abilities, allowing for involvement in clinical discussions and decision-making processes. In contrast, patients who suffer from severe stroke frequently exhibit a significant reduction in consciousness, which precludes their participation in the decision-making process [38].

Decreased consciousness, communication disorders, and cognitive impairments affect the ability of stroke patients to make decisions. Family/surrogate involvement in discussions is crucial to assist decision-making when the patient is incapacitated [39]. ACP discussions and decision-making become complex when the stroke patient does not have capacity. ACP can sometimes not be conducted between the attending physician and the patient or surrogate [33]. Therefore, implementing ACP requires an appropriate approach tailored to the patient's condition, involving family members or surrogates.

Implementing ACP in stroke patients involves collaboration among the interprofessional team to approach and respect the patient's values and wishes [40]. Nurses and other healthcare team members—such as doctors, dietitians, pharmacists, physiotherapists, occupational therapists, and speech therapists—have the potential to collaborate in delivering ACP interventions. As providers of nursing care, nurses have a role and responsibility in implementing ACP [41], as they are involved in providing information and education. They are also integral members of the healthcare team [42].

However, in this study, the role of nurses in ACP was not evident. Nurse involvement in ACP interventions has shown positive and effective results in hospital settings [43]. Nurses are considered healthcare professionals who are capable and competent to conduct ACP, as they are more familiar with the patient's family dynamics and clinical condition, have more experience in interactions and closer relationships with the patient's family, and possess strong communication skills, which are crucial in the ACP discussion process, including guiding and ensuring the patient and family make informed decisions [44]. Factors such as limited time availability, lack of competence through training or experience, lack of confidence, and discomfort with end-of-life discussions with patients hinder the involvement of healthcare professionals, including nurses, in ACP discussions [45]. Providing opportunities for ACP training, communication skills, and support will help enhance nurse involvement in ACP [45].

The implementation of ACP in the US, Europe, and Australia has progressed significantly compared to Asian countries. This is evidenced by the fact that much of the research on ACP has been conducted in the US and Europe, while ACP remains a relatively new concept in Asian countries [46]. In the US, various ACP programs have rapidly developed and proven effective [46]. However, ACP faces challenges in implementation in the US, primarily due to the lack of a unified healthcare system. This leads to high healthcare costs, limited accessibility, and inefficiencies in administration [47]. Several factors contribute to the underutilization of ACP in the US. While the public is aware of the importance of ACP, their understanding of the concept remains limited. There is also a lack of trust in the healthcare system, with individuals feeling that healthcare providers may not fully understand their wishes.

Moreover, doctors and families often believe they know what is best for the patient, so there may be no perceived need for further discussions about ACP [47]. In 2016, the government implemented a policy mandating electronic medical records (EMRs), which allowed for the documentation of ADs through EMR technology, and Medicare successfully initiated reimbursement for ACP interventions [46].

In Europe, particularly in Germany, the government supports the use of AD; however, the fragmented healthcare system poses a unique challenge to implementing ACP. Since AD became a legal part of healthcare in 2009, the quantity and quality of ADs have increased. ACP has been introduced in several nursing homes, and the necessary facilities have been provided. In 2017, a joint agreement between healthcare funders and nursing home providers was made to facilitate the implementation of ACP. ACP has been increasingly implemented in nursing homes for individuals with disabilities. To support this, various training programs have been conducted to improve the competence and qualifications of facilitators delivering ACP. However, challenges remain, including the lack of uniform qualification standards for ACP trainers, insufficient regional coordination, varying conversation content and documentation practices, and limited access to data. These factors continue to hinder the effective implementation of ACP in Germany [48].

In the Netherlands, almost all citizens are registered with a general practitioner (GP), and most ACP discussions are led by GPs. The public prefers their doctors to lead conversations regarding end-of-life care. However, for many doctors, discussing death is difficult due to limited communication skills. GPs play a key role in improving palliative care and are responsible for making decisions regarding end-of-life care. Despite ACP being studied intensively, ACP documentation is still rarely found in the records of GPs. In the Netherlands, ACP is supported by the European Association for Palliative Care (EAPC), which defines it as a process of thinking ahead, planning, and organizing—focusing on future care and current care. Training and the availability of tools are essential components developed to help raise awareness about ACP [49]. Euthanasia is included as an option in AD documentation, which is unique to the Netherlands and Belgium. In contrast, Do Not Resuscitate (DNR) orders are the most frequently documented in European countries [50].

In Asia, ACP is a relatively new concept, focusing on discussing end-of-life care and planning for future life [46], and the concept of patient autonomy is still evolving. Asian countries are diverse in culture, ethnicity, religion, and beliefs. Still, nearly all share a similar value system where the family plays a central role in decision-making, particularly regarding sick family members. Family involvement in patient care and decision-making is considered crucial. In many Asian countries, discussing death is considered taboo, as death is viewed as God's will and a natural process that every human must eventually experience [19,51]. In some Asian countries, like Singapore, Japan, and Korea, ACP has begun to be implemented through programs developed locally or adopted from the US or Europe. ACP may be initiated based on the doctor's advice, the family's wishes, or the patient's religion and beliefs [19,51].

Communication is the most important element in ACP [17] and should be considered in implementing ACP among stroke patients. ACP interventions among stroke patients have been conducted using various approaches. This study found that ACP activities can be performed by discussion, informal conversations, completing an AD, and using an ACP program. The PREPARE program is a digital program applied during ACP discussions and is complemented by media, such as AD forms, videos, pamphlets, and films [35]. Media use in ACP discussions may increase the patients' knowledge and self-efficacy in making decisions [52] and help patients and families understand the disease, treatment options, and ACP options. The risk of stroke recurrence becomes a different topic of discussion that can help patients prevent the recurrence. ACP discussions discuss the plan before death and how to live well in the future [46].

The participation of patients, families, surrogates, and healthcare professionals is a crucial strategy in implementing ACP for stroke patients. Each group is essential in ensuring that the patient's values, wishes, and preferences are respected throughout the decision-making process. Patients provide insight into their desires for care, while families and surrogates may act as decision-makers when the patient cannot communicate their wishes. Experts agree that the decision-making process during ACP may involve proxies, families, or relatives; nevertheless, those people should be trusted to make decisions on the patient's behalf [15]. Thus, families or surrogates are encouraged to be involved with ACP, especially when the patients are unable to make their own decisions [33,53].

Families' engagement during ACP benefits the patients as they can communicate what they wish to their relatives [53]. Family involvement makes ACP readiness better for the patient [54]. Including families in ACP can significantly contribute to the discussion, considering that they can provide important information, are trusted by patients, and are mainly responsible for making decisions [55]. Proxies or surrogates play a role similar to that of families who can make decisions when patients are incapacitated. In fact, given the critical roles of surrogates and proxies, they must convey their opinions; however, the accuracy of the information must be verified to ensure that what the surrogates and proxies convey is actually what the patient wishes [30,33]. A notary serves as a witness, affixing his signature to the AD document, making the AD legally binding [18]. Healthcare professionals, including doctors, nurses, and therapists, facilitate discussions, provide information, and help guide decision-making, ensuring the care plan aligns with the patient's preferences and clinical needs. This collaborative approach is vital for achieving successful and meaningful ACP outcomes in stroke care.

The implementation of ACP in stroke patients is full of challenges. The lack of information, availability of facilities, incapacitated patients, and patients' reluctance to engage in ACP discussions present unique challenges in conducting ACP for stroke patients. When a stroke patient experiences incapacity due to decreased consciousness, aphasia, severe stroke, or cognitive impairment, the family or surrogate may take over the decision-making role. A lack of information about ACP can lead to families being unwilling to participate in ACP, so it is essential to introduce ACP to patients, their families, and the broader community. A good understanding of the patient's condition increases the patient's and family's awareness of the importance of ACP. ACP discussions may help patients and healthcare providers reach a decision that can be documented in the AD and medical records. An AD is a legal document that contains health and medical care decisions and can be used when the decision-maker loses their ability to make decisions [56,57]. Evidence suggests that providing patients with clear explanations and instructions about the importance of AD would eventually lead to documentation [58].

ACP interventions can increase the patient's willingness to discuss personal values, goals of care, and desired treatments, thereby increasing AD documentation [36]. Although ACP interventions provide several benefits to the patients and their families, such practices are uncommon in stroke patients. Both patients and families have essential roles in ACP practices; however, many patients consider ACP unnecessary, likely due to their belief that their families or surrogates will provide the best

treatment for them. Thus, patients focus more on their rehabilitation than on ACP [59]. Patients benefit substantially from ADs, considering that people who communicate and document their treatment preferences would receive the medical care and comfort they prefer at the end of their lives [18].

Factors influencing ACP among stroke patients include patient condition, knowledge, and appropriate ACP methods. Age, disabilities, stroke severity [31], limited physical activities, comorbidities [37], and other conditions have been found to influence the implementation of ACP. Patients with such conditions require ACP to communicate their values, objectives, and healthcare preferences. Other factors influencing ACP practice may include history of diseases, religion and culture [60], family dynamics, family attitudes, and willingness to be involved in ACP [61]. In addition, disease knowledge, social support, educational levels, and social economy have been identified as factors contributing to ACP in stroke patients [62]. Patient knowledge, including unfamiliarity with ACP, has been discussed during ACP, including the potential risk of stroke recurrence [31,37]. Patients unaware of or do not understand their health condition will find it challenging to engage in ACP [63]. Patients with high educational levels possess better knowledge about the disease and its treatment; thus, their readiness to engage in ACP is also better. Moreover, those who receive more social support from their families and close friends are more likely to be ready for ACP discussions [62].

The PREPARE program has been widely used to conduct discussions with several patients with other health conditions [35]. Alternatively, face-to-face ACP is one approach developed based on the needs of patients with stroke and health workers. Aside from the ACP methods mentioned in this study, other ACP programs are available. This program was developed to serve as a guide when performing ACP interventions, although it has yet to be widely used in ACP interventions for stroke patients. Moreover, this program has primarily been implemented in various settings, patients, and countries—especially in developed countries—suggesting that such a program had been developed based on the needs of patients, families, and healthcare professionals.

5. Implications and limitations

One limitation of this study is the restriction in the number of articles on ACP among stroke patients that could be reviewed; hence, our findings may not have comprehensively illustrated the process of ACP among stroke patients. The results of this study showed that the practice of ACP among stroke patients is still limited, and interprofessional collaboration in stroke care has yet to show its role in the implementation of ACP. Unfortunately, nurses have yet to become involved in ACP interventions, especially in the treatment of stroke patients, highlighting the importance of increasing nurses' involvement in ACP discussions. To enhance nurses' knowledge, skills, and confidence in implementing ACP, nurses can be provided with training on ACP. Accordingly, the ACP process among stroke patients is a discussion of considerable significance. Therefore, it is necessary to develop an ACP for stroke patients method that involves interprofessional collaboration, and research on ACP in stroke patients needs to be further conducted through interprofessional collaboration.

6. Conclusions

The implementation of ACP in stroke patients is full of challenges due to various patient conditions. ACP implementation was done by conducting formal and informal conversations, completing AD documents, and using the ACP program. ACP discussion, ACP documentation, transition to CMO, and availability of surrogate decision-makers were outcomes of the studies. Effective strategies are needed to improve ACP in stroke patients. Despite the limited number of studies on ACP intervention among stroke patients, the results of this scoping review illustrated how such ACP interventions effectively increased patients' participation in ACP and improved AD documentation. In this study, family involvement in ACP was also not clearly identified. Therefore, it is important to consider further research to investigate the role of family involvement in ACP, particularly among patients with stroke.

Data availability statement

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Margareta Hesti Rahayu: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, Project administration. **Sri Sutarni**: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing - review & editing. **Heny Suseani Pangastuti**: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing - review & editing, Supervision, Project administration. **Eva Marti**: Conceptualization, Methodology, Validation, Formal analysis, Writing - review & editing.

Declaration of generative AI in scientific writing

During the preparation of this work, the author used Chat GPT in order to translate from Indonesian language to English to improve the readability and language of this manuscript.

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Declaration of competing interest

The authors declare no conflict of interest.

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Appendices. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijnss.2025.08.009.

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